Step	What We Did	Challenge	Decision & Reason
1. Data Loading	Used rul-datasets FemtoReader	Only 5 bearings loaded instead of 6, B3_2 missing from dev split	Continued with 5 clean run-to-failure bearings. Switching to raw CSVs mid-pipeline too costly. Noted as limitation.
2. EDA	Plotted raw signals, RMS trends, lifetime distribution	Bearing lifetimes ranged 1.43h–7.78h — 5.4× difference. B1_1 is 3× longer than average	Identified scale mismatch as the root cause of all previous model failures. Fix = normalise RUL per bearing.
3. Bandpass Filter	Butterworth 4th order, 2–10 kHz, zero-phase	None	Standard from Sutrisno 2012 + Cheng 2018, both validated on FEMTO. Isolates bearing fault frequencies.
4. Feature Extraction	5 features × 2 channels = 10 per window	First attempt included only RMS + Kurtosis	Added Crest Factor, Skewness, Impulse Factor based on literature — each captures independent aspect of degradation.
5. HI Construction v1	RMS-only normalised to [0,1]	98% of windows classified as Healthy — HI flat until end of life spike	RMS alone too sensitive to end-of-life spike. Needed early-fault signal.
6. HI Construction v2	PCA fusion of RMS-H + Kurtosis-H	B2_1 started at HI=0.6 — condition-dependent, physically wrong	PCA sign arbitrary, condition scaling inconsistent across operating speeds. HI demoted to visualisation only since we constructed it anyway.
7. Class Labels	First derived from HI thresholds	98% Healthy class imbalance — useless classifier	Switched to RUL-based thresholds. Guarantees consistent 25/35/25/15 split per bearing regardless of condition.
8. RUL Normalisation	RUL / max_RUL per bearing → [0,1]	Previous attempts used raw RUL cycles — scale mismatch across bearings	This single fix resolved the root cause of negative R^2 in Projects 2.0 and 2.1. Ok for this project, but real-world bearings are different.
9. Model v1 (Project 2.0)	LSTM + Attention on raw RUL targets	Negative R^2 across all bearings	Wrong data framing, not wrong model. Raw RUL without normalisation = meaningless cross-bearing targets.
10. Model v2 (Project 2.1)	CNN + SVR, temporal holdout validation	Still negative R^2, temporal holdout broken for unequal lifetimes	Temporal holdout (train first 80%, test last 20%) fails when bearing lifetimes differ 5×.
11. Validation Fix	Switched to LOBO cross-validation	Only 5 bearings — small sample	LOBO is industry standard for FEMTO. 5-fold gives 5 independent honest evaluations on unseen bearings.
12. CNN Architecture	3 conv blocks, GAP, sigmoid head	Loss not converging at 50 epochs	Increased to 100 epochs after observing loss still decreasing at ep50 in test runs.
13. SVR Predictions	CNN embeddings only → SVR	Predictions oscillating wildly window-to-window	Physically impossible — RUL can't jump 30 min in 10 seconds. Applied

			15-window rolling mean.
14. Combined Features	CNN embeddings (64d) + hand-crafted (10d) = 74d input	CNN alone R^2=0.224	Adding explicit physical features improved mean R^2 to 0.382 smoothed — CNN missing some information the hand-crafted features carry.
15. Classification	Random Forest on 10 hand-crafted features	Class imbalance 25/35/25/15	Used class_weight='balanced'. RF chosen over SVM — works on small datasets, gives feature importances, no overfitting risk.
16. B1_1 Problem	LOBO fold with B1_1 as test bearing	R^2=−0.475 / −0.608	Lifetime outlier — 7.78h vs avg 2.3h. Model trained on short bearings cannot extrapolate. Documented as known limitation (saw references.)
17. Dashboard v1	Static matplotlib — all 5 bearings, final window	All 5 badges showed Imminent Failure	This is run-to-failure dataset, all bearings died. Replaced with interactive HTML showing mid-life snapshots. Otherwise, confusing to the audience.
18. Dashboard v2	Interactive HTML, Chart.js, mid-life snapshots	Needed variety of health stages; better dashboard UX.	Auto-selected snapshot windows to show 1 Healthy, 2 Early Degradation, 1 Advanced Wear, 1 Imminent Failure — realistic fleet view.

Other versions:

Model/Version	**Challenges faced**
RF Health Classification	No RUL prediction; feature-engineering dependent; limited unseen-bearing validation.
CNN-LSTM Classification	Synthetic lifecycle labels; small dataset; overfitting risk.